Lab 39 — Model Evaluation Metrics

Aim

To compute and interpret standard classification evaluation metrics such as Confusion Matrix, Accuracy, Precision, Recall, and F1-score for a trained model.

Algorithm / Procedure

1. Load the Iris dataset.
2. Split the dataset into training and testing sets.
3. Train a Decision Tree classifier.
4. Predict the test data.
5. Generate the confusion matrix.
6. Calculate accuracy, precision, recall, and F1-score.
7. Display all evaluation metrics.

Program

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import
confusion_matrix,accuracy_score,precision_score,recall_score,f1_score

X,y=load_iris(return_X_y=True)

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=1)

clf=DecisionTreeClassifier(random_state=1)

clf.fit(X_train,y_train)

pred=clf.predict(X_test)

print("Confusion Matrix:\n",confusion_matrix(y_test,pred))

print("Accuracy:",accuracy_score(y_test,pred))
```

```
print("Precision (macro):",precision_score(y_test,pred,average='macro'))

print("Recall (macro):",recall_score(y_test,pred,average='macro'))

print("F1-score (macro):",f1_score(y_test,pred,average='macro'))
```

Output

```
Confusion Matrix:

 [[14  0  0]

 [ 0 17  1]

 [ 0  1 12]]

Accuracy: 0.9555555555555556

Precision (macro): 0.9558404558404558

Recall (macro): 0.9558404558404558

F1-score (macro): 0.9558404558404558
```

Result

The Decision Tree classifier was successfully evaluated using standard classification metrics. The model achieved approximately 95.6% accuracy, with precision, recall, and F1-score also showing strong performance.